Software Test Plan (STP)

Project: Digital Campus Library Reservation Gateway
Version: 1.0
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Revision history

Version	Date	Author	Change summary	Approval
1.0	28-10-2026	Dhareen G	First complete draft	Pending

Approvals

Role	Name	Signature / Email	Date
QA Lead			
Dev Lead			
Product Owner			

1. Introduction

Purpose: This document defines the test plan for the Digital Campus Library Reservation Gateway (LRG) v1.0. It describes objectives, scope, strategy, environment, schedule, responsibilities and traceability for testing.

Scope: Testing covers catalog search, authentication and roles, hold management, notifications and librarian administration, plus performance, reliability, accessibility, maintainability, audit and security requirements. Excluded: the internals of campus SSO and the mail service, and physical circulation hardware.

References: LRG SRS v1.0, LRG SAD v1.0, OWASP ASVS, WCAG 2.1.

Definitions: LRG (Library Reservation Gateway), SRS, SAD, RTM, UAT (User Acceptance Test), SSO, TC (Test Case), RBAC, TLS.

2. Test items

- Library Portal UI component
- Catalog Search component (including librarian catalog entry add/edit)
- Authentication component (SSO integration, sessions, roles)
- Reservation Manager component (holds, queue, return handling, demand report)
- Notification component (email)
- Library Database component (data integrity, audit log)

3. Features to be tested

Features mapped to SRS requirement IDs:

Req ID	Feature
LRG-F-001	Search catalog - The system shall allow a user to search the catalog by ISBN, title or author.
LRG-F-002	Validate search input - The system shall validate that the search term is non-empty and properly formatted (ISBN-10/13 syntax when an ISBN is entered) before querying the catalog.
LRG-F-003	Paged results + status - The system shall return results in pages showing title, author, ISBN and availability status (Available / CheckedOut).
LRG-F-004	Book detail page - The system shall display a book detail page and show a Place Hold option only when the title is CheckedOut.
LRG-F-005	No-results message - The system shall display "No results found" with advice to check spelling or broaden the search when a query matches no titles.
LRG-F-010	Campus login - The system shall authenticate members through campus credentials (single sign-on).
LRG-F-011	Guest search - The system shall allow guests to search the catalog without logging in and shall require login before a hold is placed.
LRG-F-012	Librarian role - The system shall restrict librarian functions to users holding the Head Librarian role.
LRG-F-013	Logout / idle expiry - The system shall let a user log out and shall expire a session after 30 minutes of inactivity.
LRG-F-020	Place hold - The system shall allow a logged-in member to place a hold on a CheckedOut title.
LRG-F-021	Reject invalid holds - The system shall reject a hold on an Available title and a second active hold by the same member on the same title.
LRG-F-022	View / cancel holds - The system shall let a member view and cancel their active holds.
LRG-F-023	Queue position - The system shall show a member their position in a title's hold queue.
LRG-F-024	5-hold limit - The system shall limit each member to 5 active holds.
LRG-F-030	Availability email - The system shall notify a member by email when a title they hold becomes available.
LRG-F-031	Expiry reminder - The system shall send a reminder 24 hours before a Ready hold expires.
LRG-F-040	Update availability - The system shall let the Head Librarian update a title's availability status on return or checkout, and shall move the first queued hold to Ready on return.
LRG-F-041	Hold demand report - The system shall show the Head Librarian a report of titles ranked by hold-queue length.
LRG-F-042	Manage catalog entries - The system shall let the Head Librarian add and edit catalog entries.
LRG-NF-001	Search response time - A catalog search shall return its first page within 2 seconds for 90% of requests with 200 concurrent users.
LRG-NF-002	Availability 99.5% - The system shall be available 99.5% of each month during the academic term, excluding scheduled maintenance.
LRG-NF-003	Data confidentiality - Member identity and hold history shall be visible only to the member concerned and the Head Librarian.
LRG-NF-004	Accessibility - The portal shall meet WCAG 2.1 AA and work on current Chrome, Firefox and Edge and on mobile browsers.
LRG-NF-005	Scalability - The system shall support 5,000 registered members and 100,000 catalog titles without exceeding the LRG-NF-001 search time.

Req ID	Feature
LRG-NF-006	Maintainability - Components shall communicate only through documented interfaces, and business-layer code shall have at least 70% unit-test coverage.
LRG-NF-007	Audit logging - Hold creation, hold cancellation and librarian status changes shall be logged with user, time and action and retained for 1 year.
LRG-SR-001	TLS 1.2+ - TLS 1.2 or higher shall be mandatory for all connections between browser, application, SSO provider and mail server.
LRG-SR-002	No stored passwords - The system shall never store or log member passwords; authentication shall rely on campus SSO tokens.
LRG-SR-003	Server-side authorization - Every hold and librarian operation shall be authorized on the server by role and ownership.
LRG-SR-004	Input validation / SQL - All user input shall be validated and all database access shall use parameterized queries.
LRG-SR-005	Login lockout - After 5 failed logins within 15 minutes, further attempts from that account or address shall be blocked for 15 minutes and the event logged.
LRG-SR-006	Cookie flags - Session cookies shall be marked Secure, HttpOnly and SameSite=Strict.
LRG-SR-007	Rate limiting - Search and API endpoints shall be rate-limited to 60 requests per minute per client.

4. Features not to be tested

- Campus SSO provider internals (assumed tested by the institution); SSO is replaced by a stub in most tests.
- Mail server delivery internals; a test SMTP sink (e.g. MailHog) is used.
- Physical check-out hardware and fines/payments (not part of v1.0).
- Browser vendor bugs outside the supported browser list.

5. Test approach / strategy

Levels:

- Unit tests (business-layer classes, validators, queue logic) - JUnit, target 70% coverage (LRG-NF-006)
- Integration tests (component to component through interfaces, and with the database and stubs)
- System tests (end-to-end flows UC-01 to UC-09 through the UI and API)
- Acceptance tests (UAT by the Head Librarian and sample students)

Types:

- Functional testing (Appendix A, TC-SRCH, TC-AUTH, TC-HOLD, TC-NOTF, TC-ADM)
- Regression testing (automated suite re-run after every fix and before UAT)
- Performance testing (TC-PERF-01, TC-PERF-02 with JMeter)
- Usability and accessibility testing (TC-UX-01)
- Reliability and maintainability checks (TC-AVL-01, TC-MNT-01)

Entry criteria: Stable build deployed to the test environment, test data loaded, SSO and SMTP stubs running, smoke test passed.

Exit criteria: 100% of planned test cases executed, 0 Critical and 0 High defects open, at least 95% of test cases passed, all acceptance criteria in the SRS satisfied, RTM updated.

5.1 Security validation

- Verify TLS 1.2+ only and HTTP redirect (TC-SEC-01)
- SQL injection and script (XSS) input testing on search and forms (TC-SEC-02)
- Cookie flags Secure, HttpOnly, SameSite (TC-SEC-03)
- Access-control tests: other member's data and librarian endpoints with a Student token (TC-SEC-04, TC-SEC-06, TC-AUTH-04)
- Check that passwords never appear in the database or logs (TC-SEC-05)
- Login lockout and rate limiting (TC-AUTH-02, TC-SEC-07)
- Optional: OWASP ZAP baseline scan of the test deployment

6. Test environment

Item	Details
Hardware	Laptop or cloud VM (2 vCPU, 4 GB RAM) for server; test laptops and one Android phone for clients
Software	LRG v1.0 build, PostgreSQL/MySQL test database, SSO stub (mock OpenID provider), MailHog SMTP sink, Chrome, Firefox, Edge
Tools	Selenium (UI automation), Postman (API), JMeter (performance), JUnit (unit), OWASP ZAP (security scan), Lighthouse/axe (accessibility), Jira (defect tracking), GitHub (test artefacts)
Test data	100,000 generated titles (a subset with CheckedOut status), 5,000 dummy members, one Head Librarian account, accounts with 0, 4 and 5 active holds

7. Test schedule

Proposed dates; adjust to the course calendar.

Milestone	Date
Test case design	30-Oct-2026
Environment setup	02-Nov-2026
Test execution start	03-Nov-2026
Test execution end	13-Nov-2026
Regression run and defect closure	14-Nov-2026 to 15-Nov-2026
UAT	16-Nov-2026 to 18-Nov-2026
Test Summary Report	19-Nov-2026

8. Test deliverables

- Test Plan (this document)
- Test cases (Appendix A; manual and automated)
- Test scripts (Selenium, JMeter, Postman collections)
- Test data sets
- Test execution logs
- Defect reports (Jira)
- Test Summary Report

9. Roles and responsibilities

Role	Name	Responsibility
QA Lead	Dhareen G	Prepare plan, coordinate execution, report status
Test Engineer	<Name>	Design and execute test cases, log defects
Developer	<Name>	Support defect fixes and triage, write unit tests
Product Owner	<Name>	Approve test results, sign off readiness

10. Risks and mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds from developers; start API tests on the first stable endpoints
Test environment downtime	Keep a backup environment on a cloud VM and scripted setup
SSO or email service unavailable	Use SSO and SMTP stubs for all tests except a final integration check
Load-test results not representative	Seed production-sized data and run on hardware similar to deployment
Few team members for testing	Prioritize High-priority test cases; automate regression

11. Assumptions and dependencies

- Test data will be loaded before execution starts.
- SSO stub and SMTP sink are available in the test environment.
- Requirements in SRS v1.0 are frozen during the test cycle; changes follow change control and trigger updates to the RTM.
- The Head Librarian is available for UAT.

12. Suspension and resumption criteria

Suspend testing if:

- Test environment is unavailable for more than 4 hours
- Build is too unstable (blocks more than 30% of test cases)
- A Critical defect (data loss, security breach) is found, until fixed

Resume testing if:

- Blocking defects are resolved and verified
- Environment is stable and a new smoke test passes

13. Test case management and traceability

The RTM maps each SRS requirement to its test cases. Full list below; test case details are in Appendix A.

Req ID	Requirement	Test case(s)
LRG-F-001	Search catalog	TC-SRCH-01, TC-SRCH-02
LRG-F-002	Validate search input	TC-SRCH-03
LRG-F-003	Paged results + status	TC-SRCH-04
LRG-F-004	Book detail page	TC-SRCH-05
LRG-F-005	No-results message	TC-SRCH-06
LRG-F-010	Campus login	TC-AUTH-01, TC-AUTH-02
LRG-F-011	Guest search	TC-AUTH-03
LRG-F-012	Librarian role	TC-AUTH-04
LRG-F-013	Logout / idle expiry	TC-AUTH-05
LRG-F-020	Place hold	TC-HOLD-01
LRG-F-021	Reject invalid holds	TC-HOLD-02
LRG-F-022	View / cancel holds	TC-HOLD-03
LRG-F-023	Queue position	TC-HOLD-04
LRG-F-024	5-hold limit	TC-HOLD-05
LRG-F-030	Availability email	TC-NOTF-01
LRG-F-031	Expiry reminder	TC-NOTF-02
LRG-F-040	Update availability	TC-ADM-01
LRG-F-041	Hold demand report	TC-ADM-02
LRG-F-042	Manage catalog entries	TC-ADM-03
LRG-NF-001	Search response time	TC-PERF-01
LRG-NF-002	Availability 99.5%	TC-AVL-01
LRG-NF-003	Data confidentiality	TC-SEC-04
LRG-NF-004	Accessibility	TC-UX-01
LRG-NF-005	Scalability	TC-PERF-02
LRG-NF-006	Maintainability	TC-MNT-01
LRG-NF-007	Audit logging	TC-AUD-01
LRG-SR-001	TLS 1.2+	TC-SEC-01
LRG-SR-002	No stored passwords	TC-SEC-05
LRG-SR-003	Server-side authorization	TC-AUTH-04, TC-SEC-06
LRG-SR-004	Input validation / SQL	TC-SEC-02
LRG-SR-005	Login lockout	TC-AUTH-02
LRG-SR-006	Cookie flags	TC-SEC-03
LRG-SR-007	Rate limiting	TC-SEC-07

Defects are tracked in Jira with severity (Critical, High, Medium, Low) and linked to the failing test case ID. Test status is recorded as Not Run, Pass, Fail or Blocked.

14. Test metrics and reporting

- % test cases executed
- % passed / failed / blocked
- Defect density (defects per requirement or per module)
- Defect aging (days open)
- Requirement coverage (requirements with at least one passed test)

Reports: daily execution status; final Test Summary Report.

15. Approvals

Role	Name	Signature / Date
QA Lead		
Dev Lead		
Product Owner		

Appendix A: Test case catalogue

TC ID	Objective	Requirements	Type	Priority	Expected result
TC-SRCH-01	Search by a valid ISBN	LRG-F-001	Functional	High	The matching title is returned with its availability.
TC-SRCH-02	Search by partial title and by author	LRG-F-001	Functional	High	All titles containing the term or by that author are listed.
TC-SRCH-03	Submit empty and malformed search terms	LRG-F-002	Functional	High	A validation message is shown and no database query runs.
TC-SRCH-04	Check pagination and result fields	LRG-F-003	Functional	High	10 results per page with title, author, ISBN and status; paging works.
TC-SRCH-05	Open details for Available and CheckedOut titles	LRG-F-004	Functional	Medium	Place Hold appears only on the CheckedOut title.
TC-SRCH-06	Search for a term with no matches	LRG-F-005	Functional	Low	"No results found" with guidance; no list shown.
TC-AUTH-01	Log in with valid campus credentials	LRG-F-010	Functional	High	Session is created and the member name is shown.
TC-AUTH-02	Log in with invalid credentials 6 times	LRG-F-010, LRG-SR-005	Security	High	Errors shown, no session; 6th attempt blocked and logged.
TC-AUTH-03	Guest searches, then clicks Place Hold	LRG-F-011	Functional	Medium	Search works as guest; Place Hold redirects to login.
TC-AUTH-04	Student attempts librarian functions	LRG-F-012, LRG-SR-003	Security	High	Access denied (403) and the attempt is logged.
TC-AUTH-05	Logout and 30-minute idle expiry	LRG-F-013	Functional	Medium	Protected pages require login again.
TC-HOLD-01	Place a hold on a CheckedOut title	LRG-F-020	Functional	High	Hold created; confirmation shows queue position.
TC-HOLD-02	Hold on an Available title; duplicate hold	LRG-F-021	Functional	High	Both rejected with a clear reason; no record created.
TC-HOLD-03	View and cancel an active hold	LRG-F-022	Functional	Medium	Hold removed; queue positions update.
TC-HOLD-04	Check queue position after a cancellation	LRG-F-023	Functional	Low	Positions behind the cancelled hold move up by one.
TC-HOLD-05	Request a sixth active hold	LRG-F-024	Functional	Medium	Request rejected with a limit message.
TC-NOTF-01	Return a held title	LRG-F-030	Functional	Medium	Email naming the title reaches the first queued member.
TC-NOTF-02	Reminder 24 hours before hold expiry	LRG-F-031	Functional	Low	Reminder sent once; none for cancelled holds.
TC-ADM-01	Librarian marks a title returned	LRG-F-040	Functional	High	Status Available; first hold becomes Ready.
TC-ADM-02	Open the hold demand report	LRG-F-041	Functional	Low	Titles ranked by queue length; counts match the database.
TC-ADM-03	Add and edit a catalog entry	LRG-F-042	Functional	Medium	Entry saved and searchable.
TC-PERF-01	200 concurrent users run searches	LRG-NF-001	Performance	High	90th percentile response <= 2 s.
TC-PERF-02	Search with 100,000 titles and 5,000 members seeded	LRG-NF-005	Performance	Medium	Search time stays within LRG-NF-001.
TC-AVL-01	Review one month of uptime monitoring	LRG-NF-002	Reliability	High	Availability >= 99.5% excluding maintenance.
TC-UX-01	Accessibility and responsive-layout audit	LRG-NF-004	Usability	Medium	WCAG 2.1 AA checks pass; layout works at 360 px.
TC-MNT-01	Review coverage and interface usage	LRG-NF-006	Maintainability	Medium	Coverage >= 70%; no direct UI-to-database calls.
TC-AUD-01	Perform holds and status changes, then read the logs	LRG-NF-007	Functional	High	Each action has a log entry with user, time and action.
TC-SEC-01	Scan TLS configuration	LRG-SR-001	Security	High	Only TLS 1.2/1.3 accepted; HTTP redirects to HTTPS.
TC-SEC-02	Submit SQL injection and script payloads	LRG-SR-004	Security	High	Payloads rejected or neutralized; no data leaks.
TC-SEC-03	Inspect session cookie attributes	LRG-SR-006	Security	Medium	Secure, HttpOnly and SameSite=Strict present.
TC-SEC-04	Request another member's hold data	LRG-NF-003	Security	High	Response is 403; no foreign data returned.
TC-SEC-05	Inspect database and logs for passwords	LRG-SR-002	Security	High	No passwords stored or logged.
TC-SEC-06	Call librarian endpoints with a Student token	LRG-SR-003	Security	High	403 returned for every librarian endpoint.
TC-SEC-07	Send 61 requests in one minute	LRG-SR-007	Security	Medium	Request 61 returns HTTP 429.